IHFC Global Heat Flow Database - Data Structure | Metadata | Quality Code

Use the videos below to get an quick introduction into the data and meta data structure and the quality scheme
of the Global Heat Flow Database of the International Heat Flow Commission (IHFC).

The data structure explained

Core Metadata Elements of the Database

A key feature of the portal is the inclusion of persistent identifiers to the machine-readable heat flow metadata that support citability and ensure traceability and data integration,:

DOI - Digital Object Identifier

Assign DOI (Digital Object Identifier) to newly submitted heat flow data and regular publications of the entire database content to ensure their long-term citability and accessibility, together with the provision of rich metadata for data discovery.

ORCID - Open Researcher and Contributor ID

ORCID (Open Researcher and Contributor ID) are used to uniquely identify researchers contributing to the database.

IGSN - International Generic Sample Number

The IGSN (International Generic Sample Number) links datasets and publications to the physical samples on which the measurements were performed.

ROR - Research Organization Registry

The ROR (Research Organization Registry) is a community-led registry of open persistent identifiers for research organizations.

Data and Metadata Structure

The IHFC database structure shows associated data fields for the parent and child level [Level] relevant for heat flow density determinations [Domains] from boreholes and mines (B) and for shallow probe-sensing data (S). Database fields [Obligation] are classified as mandatory (M), recommended (R), or optional (O). The relevance for the quality evaluation [Quality] is displayed for the U-score (U), the M-score (M), and the perturbation effects (P).

ID	Field name	Category	Domain	Obligation	Quality	Level
P01			B,S	M	U-score (B,S)	Parent
P02	Heat-flow uncertainty		B,S	M	U-score (B,S)	Parent
P03	Site name		B,S	M		Parent
P04	Latitude (Geographical)		B,S	M		Parent
P05	Longitude (Geographical)		B,S	M		Parent
P06	Elevation (Geographical)		B,S	M	M-score (S)	Parent
P07	Basic geographical environment		B,S	M		Parent
P08	General comments parent level		B,S	R		Parent
P09	Flag heat production of the overburden		B,S	R		Parent
P10	Total measured depth		B	R		Parent
P11	Total true vertical depth		B	R		Parent
P12	Type of exploration method		B	M		Parent
P13	Original exploration purpose		B	R		Parent
C01		Heat flow	B,S	M	U-score (B,S)	Child
C02	Heat-flow uncertainty child	Heat flow	B,S	M	U-score (B,S)	Child
C03	Heat-flow method	Heat flow	B,S	M		Child
C04	Heat-flow interval top	Heat flow	B,S	M	M-score (B, S)	Child
C05	Heat-flow interval bottom	Heat flow	B	M	M-score (B)	Child
C06	Penetration depth	Heat flow	S	M	M-score (S)	Child
C07	Primary publication reference	Meta data	B,S	M		Child
C08	Primary data reference	Meta data	B,S	R		Child
C09	Relevant child	Meta data	B,S	M		Child
C10	General comments child level	Meta data	B,S	R		Child
C11	Flag in-situ thermal properties	Meta data	B,S	R		Child
C12	Flag temperature corrections	Meta data	B,S	M	M-score (S)	Child
C13	Flag sedimentation effect	Meta data	B,S	M	P-flag	Child
C14	Flag erosion effect	Meta data	B,S	M	P-flag	Child
C15	Flag topographic effect	Meta data	B,S	M	P-flag	Child
C16	Flag paleoclimatic effect	Meta data	B,S	M	P-flag	Child
C17	Flag surface temperature/bottom water	Meta data	B,S	M	P-flag	Child
C18	Flag convection processes	Meta data	B,S	M	P-flag	Child
C19	Flag heat refraction effect	Meta data	B,S	M	P-flag	Child
C20	Expeditions/Platforms/Ship	Meta data	B,S	R		Child
C21	Probe type	Meta data	S	R		Child
C22	Probe length	Meta data	S	R		Child
C23	Probe tilt	Meta data	S	M	M-score (S)	Child
C24	Bottom-water temperature	Meta data	S	O		Child
C25	Lithology	Meta data	B,S	O		Child
C26	Stratigraphic age	Meta data	B,S	O		Child
C27		Gradient	B,S	M		Child
C28	Temperature gradient uncertainty	Gradient	B,S	R		Child
C29	Mean temperature gradient corrected	Gradient	B,S	O		Child
C30	Corrected temperature gradient uncertainty	Gradient	B,S	O		Child
C31	Temperature method (top)	Gradient	B	M	M-score (B)	Child
C32	Temperature method (bottom)	Gradient	B	M	M-score (B)	Child
C33	Shut-in time (top)	Gradient	B	R		Child
C34	Shut-in time (bottom)	Gradient	B	R		Child
C35	Temperature correction method (top)	Gradient	B	R		Child
C36	Temperature correction method (bottom)	Gradient	B	R		Child
C37	Number of temperature recordings	Gradient	B,S	M	M-score (B, S)	Child
C38	Date of acquisition	Gradient	B,S	M		Child
C39	Mean thermal conductivity	TC	B,S	M		Child
C40	Thermal conductivity uncertainty	TC	B,S	R		Child
C41	Thermal conductivity source	TC	B,S	M	M-score (B, S)	Child
C42	Thermal conductivity location	TC	B,S	M	M-score (B, S)	Child
C43	Thermal conductivity method	TC	B,S	M	M-score (S)	Child
C44	Thermal conductivity saturation	TC	B,S	M	M-score (B, S)	Child
C45	Thermal conductivity pT conditions	TC	B,S	M	M-score (B, S)	Child
C46	Thermal conductivity pT assumed function	TC	B,S	R		Child
C47	Thermal conductivity number	TC	B,S	M	M-score (B, S)	Child
C48		TC	B,S	R		Child
C49	IGSN	TC	B,S	O		Child
A1	Reviewer name		x			Admin
A2	Reviewer comment		x			Admin
A3	Review date		x			Admin
A4	Country		x			Admin
A5	Region		x			Admin
A6	Continent		x			Admin
A7	Domain		x			Admin
A8	Unique entry ID		x			Admin

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